TRAIN SIM CLASSIC · VEHICLE WORKFLOW GUIDE

Part 4

Blueprint Editor 2: turning geometry into a vehicle

Build the bogie, simulation and vehicle blueprints in the right order, understand every field that matters, and avoid the unit and default-value traps the sample vehicles contain.

In this part

1. The blueprints you need and the order to build them
2. How BPE2 stores values (and why unit labels lie)
3. Bogie blueprints
4. The engine simulation blueprint: traction, brakes, air, curves
5. The engine blueprint: render, rail vehicle, children
6. Controls and the cab
7. Wagons and tenders
8. Exporting and previewing
9. Field reference: sample values side by side

All field names and sample values are read from the Kuju sample blueprints supplied with this project unless marked **check**.

1 · The blueprints you need and the order to build them

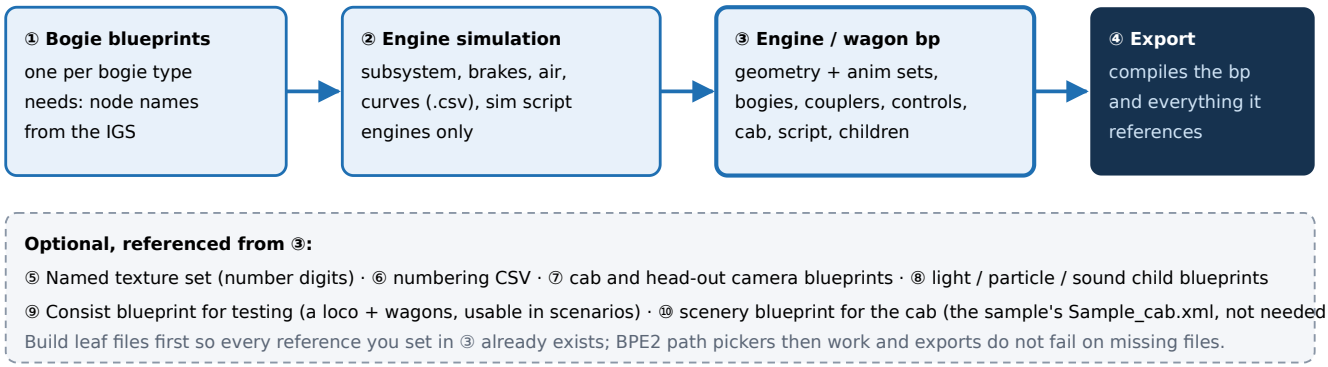


Figure 4.1 · Build order. Leaves first, root last.

> **Start from a copy, not a blank**
>
> Copy the Kuju sample blueprints into your product folder and rename them. A blank blueprint has dozens of fields with no obvious defaults; a working sample already has valid brake, air and control structures. Then change what you understand and verify the rest. Keep the `Developer\Addon` provider/product in references only if you really use that folder; otherwise re-point every reference.

2 · How BPE2 stores values (and why unit labels lie)

A value typed in BPE2 is stored twice: the real `<Value>` the game reads, and a `ToolData` block recording what you typed (`mTypedIn`) and the unit label shown (`mUnits`). Some files have no ToolData at all (the sample engine and Bogie 01) and are equally valid. **verified**

```
<Attribute name="MaxForce"><sFloat32><Element>
  <ToolData> … mUnits = "Lb/f"  mTypedIn = "35.296001434326172" … </ToolData>
  <Value>35.296001</Value>          <- the number the game uses
</Element></sFloat32></Attribute>
```

Label seen in samples	Observation	Advice
7F `MaxForce` “Lb/f” = 35.296	The real SDJR 7F tractive effort is 35,296 lbf, so the value is in thousands of lbf despite the label.	Compare with a known-good loco
Electric `MaxPower` / `MaxForce` no label = 6600 / 300	Throttle curve peaks at exactly 300, consistent with kN.	Keep curve peak = MaxForce
Handbrake `MaxForce` “Lb/f” = 10000 / 15000	The handoff doc treats this as newtons; the label says lbf.	Tune in game; watch for locked wheels

Label seen in samples	Observation	Advice
Compressor `PowerRating` “PSI_PER_SECOND” = 3	Label is obviously wrong for a power rating.	Ignore the label
Air `ReservoirVolume` “PSI” = 10	Label is obviously wrong for a volume.	Ignore the label

Boolean fields use `eTrue` / `eFalse`; enums are strings (`eVacuumSinglePipe`, `eDoubleCab`). Lists are wrapped in `__Indexed__…` elements; an empty list is `<Element><Value/></Element>`.

3 · Bogie blueprints

Field	Sample (Bogie 01)	Meaning
`WheelRadius`	0.55	Metres. Must equal the modelled wheel radius (mesh: 0.525 ✘).
`WheelGauge`	1.435	Metres; standard gauge.
`Geometry`	`bo01`	Node name of the bogie frame in the vehicle IGS.
`Animation`	(empty)	Optional animation on the bogie.
`Powered`	eTrue	Whether this bogie's axles deliver traction. Bogie 02 omits it (engine default). Set it explicitly.
`SuspensionStiffness` / `Damping`	8E+09 / 6E+07	Very stiff = rigid. Bogie 02 omits both. Set them explicitly.
`Axle[]`: Radius, HorizontalOffset, NodeId, AnimationId	0.55, ±2.2 / 0, `bo01wh01..03`	One entry per wheelset; offset along track from the bogie pivot.
`PivotOffset`	0 / 0	Extra horizontal/vertical pivot shift.

> **Let Blender write these**
>
> The toolkit's **Measure + write bogie XML** generates a bogie blueprint per `boNN` object with radius, offsets and node names measured from the mesh, in the same structure as Kuju's Bogie 01. Point the engine blueprint at the generated files.

4 · The engine simulation blueprint

4.1 Common subsystem fields

Field	Electric sample	Steam 7F	Notes
`MaxPower`	6600	1328 (hp label)	Hard cap on power.
`MaxForce`	300	35.296	Hard cap on tractive effort; ≥ curve peaks.
`MaxContinuousForce`	250	–	≤ MaxForce.

Field	Electric sample	Steam 7F	Notes
`MaxSpeed`	145 mph	60 mph	≥ the vehicle's MaxPermissibleSpeed.
`StartupTime` / `ShutdownTime`	5 / 2 s	10 / 15 s	Time from Startup control to ready.
`Has3rdRailPowerSupply`	eFalse	eFalse	Third-rail pickup.

4.2 Electric-specific fields (sample)

`HasPantograph` eTrue · `TractiveForceVSpeed` and `TractiveEffortVThrottle` CSVs · `HasThyristorControl` eTrue · `OverloadCurrent` 650 A · `MaxCurrent` 1500 A · `AntiSlipBrakeCylinderPressure` 20 psi · `CylinderPressureForBrakeToCutPower` 30 psi · `QuickPowerOffAtZeroThrottle` eFalse · `SlowSpeedFitted…` · `HasVentilator` eFalse · `DynamicBrakesSystem` (delay 2 s, 3–100 mph, max force 150, resistor 1000 A, `BrakeEffortVsSpeed` CSV). **verified**

4.3 Steam-specific blocks (7F sample)

Firebox (hand-fired, 800–1700 °C, 1651 kg max coal, grate limit 3080 lb/h) · **Boiler** (200 psi, 210 ft³, 3900 kg water, superheater 1.22, two safety valves 187/190 psi) · **Cylinders** (2 × 21″×28″, manual cocks) · **Regulator** (twin-port) · **Injectors** (live and exhaust) · **Blower** · **CompressorEjector** (small and large ejectors) · **SteamHeating** · curves `TractiveEffortVsCutoff` (0–1 multiplier) and `TractiveEffortLinearVsSpeed` · `TenderRequired` eTrue. **verified**

4.4 Brakes

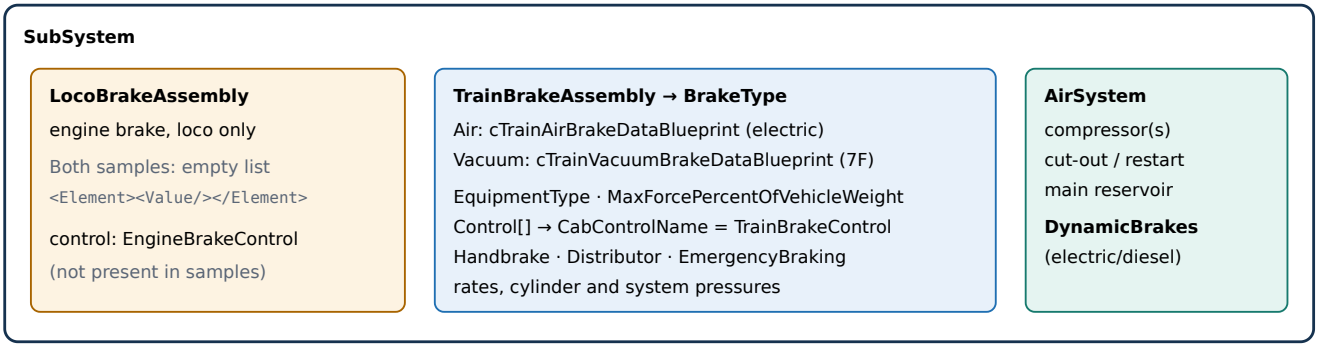


Figure 4.2 · Brake structure inside the subsystem. The only link to the cab is `CabControlName`, which must equal a `ControlName` in the vehicle blueprint.

> **The LocoBrakeAssembly question**
>
> The project's handoff document states that an empty `<LocoBrakeAssembly />` crashes newer TSC builds. Both Kuju samples ship with an *empty indexed list* (`<Element><Value/></Element>` inside the attribute), which is a different XML shape. Safe practice: never hand-write a self-closing `LocoBrakeAssembly`; if your build is unstable or you want an independent brake, populate a loco-only brake that mirrors the train brake and add an `EngineBrakeControl`. **check** the crash claim on your build.

Train brake field	Electric (air)	7F (vacuum)	Notes
`EquipmentType`	eAirTwinPipe	eVacuumSinglePipe	Must match the wagons you haul.
`MaxForcePercentOfVehicleWeight`	65	60	Brake force as % of weight.
`MaxApplicationRate` / `MaxReleaseRate`	10 / 10 psi/s	5 / 1 inHg/s	How fast brakes bite / release.
`MaxCylinderPressure`	63 psi	21 inHg	Peak cylinder pressure.
`PressureForMaxForce`	63	**0**	Pressure at which full force is reached. The 7F's 0 looks suspicious; keep it equal to max cylinder pressure unless you know otherwise.
`MaxSystemPressure`	72.5 psi	21 inHg	Brake-pipe working pressure.
`GraduatedRelease`, `ProportionalBrake`	eTrue, eTrue	–, eTrue	Partial release allowed; force ∝ pressure.
Control: `EmergencyApplicationRate`	55	55	Emergency application speed.
Control: `FullServicePressureDrop`	21.5 psi	15	Pipe drop for full service.
`Handbrake.MaxForce`	15000	10000	See unit note in §2.

Air system (electric): cut-out 145 psi, restart 123 psi, main reservoir max 145 psi, one electric compressor. A `CutOutPressure` of 0 or an empty compressor list means the brakes never charge; the 1890s trolley has both.

4.5 Curves (.csv)

Plain two-column files, no header, `x,y` per line, linearly interpolated **verified**:

```
TractiveEffortVsThrottle.csv (electric)
0,0
10,30
20,60
…
100,300
```

```
TractiveEffortVsCutoff.csv (7F steam)
0,0.18
0.1,0.43
0.2,0.53
…
0.75,1
```

- Throttle curve: X = throttle in %, Y = force; start at 0,0; peak = MaxForce.
- Speed curve: X = speed, Y = max force; last point at or beyond MaxSpeed.
- Steam cutoff curve: X = cutoff fraction, Y = multiplier 0–1.

5 · The engine blueprint

5.1 RenderComponent

- `GeometryID`: the exterior IGS path (relative to the product, e.g. `RailVehicles\Electric\EngineSample\Default\Engine\Sample_Engine t3.igs`).

- `CollisionGeometryID` : a simple hull IGS (the sample uses a separate file).
- `PrimaryNamedTextureSet` / `Secondary…` : the digits texture set used by `primarydigits` .
- `AnimSet` : one entry per exterior .ia: `AnimationID` (your name), `AnimationName` (path), `AnimateInEditor` .
- `ShadowType` , `CastsShadows` , `DetailLevelGenerationRange` : leave as the sample unless you know why.

5.2 RailVehicleComponent

Field	Electric	Steam 7F	Guidance
`Mass` (t)	84	68	Real service mass; it sets adhesion and brake force.
`EaseOfDerailment`	0.4375	0.4375	Lower = harder to derail.
`FrontPivotX` / `BackPivotX`	±9.515	6.6 / −4.185	Body pivots, see Part 1.
`Collision…`	18 × 2.978 × 2.75	10.8 × 2.5 × 3.0	Generate from the mesh (toolkit).
`DragCoefficient`	1.45	2.04	Air resistance.
`RollingFrictionCoefficient`	0.0006	0.00065	Rolling resistance.
`Dry/Wet/SnowFriction`	0.9 / 0.45 / 0.275	1.0 / 0.5 / –	Adhesion by weather.
`SandFrictionMultiplier`	2	2	Effect of sanding.
`Bogey[]`	+6.25, −5.583	+5.151, 0	Must match the mesh (toolkit patch).
`Front/RearCouplingBlueprintID`	**empty**	screw / bar coupling	Set one; the electric leaves both empty.
Coupling pivot / receiving matrices	±9.03 / ±9.14 at 1.025	–	From empties in Blender (toolkit).
`EngineType`	Electric	Steam	HUD and default key bindings.
`MaxPermissibleSpeed`	**0**	–	Set to the real limit; AI drives to it.
`ValidStandaloneLoco`	eTrue	–	Allows use as a lone loco.
`NumberingList`	Sample_numbers.csv	Example_Steam_numbers.csv	Running numbers for the digits.

5.3 Children (ContainerComponent)

Child entities attach other blueprints at a position: head/tail light sources, particle emitters (the 7F's `Main Smoke Stack` → `Particles\Example_Smoke.xml`), sound emitters. The **ChildName** is how Lua reaches it (`Call("Main Smoke Stack:SetEmitterRate", r)`). The sample electric has no children. verified

6 · Controls and the cab

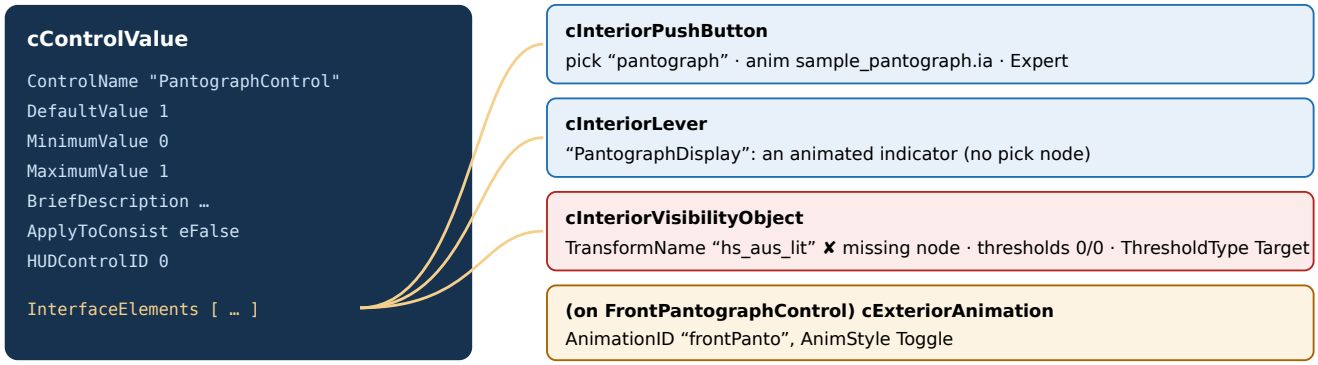


Figure 4.3 · One control value, several interface elements. The sample's `PantographControl` has a clickable button, a display lever and a lamp; a lamp whose node is missing silently never shows. verified

Element type	Key fields	Typical use
`cInteriorLever`	PickTransformName, AnimationName, MovementType, sensitivities	Throttle, needles (no pick)
`cInteriorNotchedLever`	+ NumberOfNotches, RestInNotchedPosition	Reverser, headlight switch
`cInteriorIrregularNotchedLever`	+ Notch[] (Identifier, Value), DisplayNotchName	Brake handles
`cInteriorPushButton`	+ AutoRelease	Horn, start, pantograph
`cExteriorAnimation`	AnimationID, AnimStyle (Toggle, BackAndForthContinuous…)	Pantograph, wipers, doors
`cInteriorGeometrySlaveOutput`	AnimationName	Second cab animation from same control
`cInteriorVisibilityObject`	TransformName, Lower/UpperVisibilityThreshold, ThresholdType	Lamps, indicators, mesh swaps

MovementType (MoveUpDown, MoveLeftRight, MoveRightLeft, MoveDownUp, PushButton) sets how the mouse drags the control. **DifficultyType** (StopGo, Intermediate, Expert) chooses which elements are active per difficulty. The sample brake handle's notches are Release 0, GraduatedSelfLapLimitedHolding 0.3, Suppression 0.85, ContinuousService 0.9, Emergency 0.95. verified

Control names found in the samples

Group	Names
Driving	`Regulator`, `Reverser`, `SimpleThrottle`, `SimpleChangeDirection`, `TrainBrakeControl`, `EmergencyBrake`, `HandBrake`, `Startup`, `EngineStart`
Electric	`PantographControl`, `FrontPantographControl`, `RearPantographControl`
Steam	`FireboxDoor`, `Stoking`, `Blower`, `Damper`, `CylinderCock`, `LiveInjectorSteamOnOff`, `LiveInjectorWater`, `ExhaustInjectorSteamOnOff`, `ExhaustInjectorWater`, `SmallCompressorOnOff`, `WaterScoopRaiseLower`, `SafetyValve1/2`
Ancillary	`Horn`, `Sander`, `Headlights`, `Wipers`, `DoorsOpenClose`
Gauges (outputs)	`SpeedometerMPH`, `BoilerPressureGaugePSI`, `SteamChestPressureGaugePSI`, `VacuumBrakePipePressureINCHES`, `SteamHeatingPressureGaugePSI`, `WaterGauge`

The simulation recognises its inputs and outputs by these exact names **check** which ones your subsystem reads before renaming anything. Your own custom names (e.g. `TrolleyPoleFront`) are ignored by the simulation and are free for scripts and animations.

The cab (InteriorInterface)

`InteriorGeometryID` = cab IGS, `NumberOfCabs` = eSingleCab/eDoubleCab, `InteriorCamera` = cab camera blueprint, `CabOcclusion` optional. The head-out camera and `DriverPosition` matrix live on the rail vehicle component.

7 · Wagons and tenders

A wagon is built exactly like an engine minus the powertrain. **check** exact class and field names against a stock wagon in your BPE2 (New ▸ Wagon blueprint), since no wagon sample was supplied:

- Same Blender rules: `bo01` / `bo02`, wheels, origin on the rail, LODs, shadows, numbers.
- Rail vehicle data: mass (empty and loaded if the wagon carries cargo), friction, bogies, couplings, collision.
- Brake equipment must match the loco (air twin pipe vs vacuum), or the train brake does not propagate.
- No engine simulation, no cab. Controls only if a script or animation needs them (doors, tail lamps).
- Scripts are allowed: doors, loads and lamps are typically driven by `OnConsistMessage` from the loco (Part 5).
- Tenders are their own type; a steam engine with `TenderRequired = eTrue` needs one coupled to run.

8 · Exporting and previewing

1. Save every blueprint (BPE2 validates as you save and highlights broken references).
2. Right-click the vehicle blueprint (or its folder) ▸ **Export**. BPE2 compiles the blueprint and everything it references into `Assets\Provider\Product\…`: geometry, textures, animations, curves and Lua (`.out`). Read the output pane for errors.

3. Use **Preview** to open the compiled vehicle in the asset preview, check textures, LODs and "Animate in editor" animations. If an animation looks stale, close the preview and clear its cache (BRIAGE §9.1.1).
4. Test in game (Part 6).

> **Never export into a route that is open**
>
> Exporting to an enabled provider/product while the TSC route editor has it loaded, then reloading the product, can corrupt the route on save (BRIAGE §9.1.3). Close the game before exporting.

9 · Field reference: problems in the samples and the 1890s trolley

Symptom in game	Field(s)	Fix
Spawns with brakes on	`TrainBrakeControl.DefaultValue` 0.7 / 1.0 / 0.25	0 for driving tests
AI tries to go too fast / no limit	`MaxPermissibleSpeed` 0 or 100 on a 35 km/h tram	Real limit
Almost no traction	`MaxForce` 10 vs curve peak 220 (trolley)	Match curve and cap
Brakes never charge	Empty compressor, `CutOutPressure` 0 (trolley)	Add compressor, set pressures
Brake handle does nothing	`CabControlName` ≠ ControlName	Same string
Wheels lock on handbrake	`Handbrake.MaxForce` 100000 (trolley)	Tune down
Bogie turns in the wrong place	`BogeyPivotX` ≠ mesh	Toolkit patch
Couplers don't connect / float	Empty coupling blueprint IDs; matrices off the ends	Set coupling blueprints; toolkit patch

> **Automate the lint**
>
> `python tools/tsc_crosscheck.py --engine … --sim … --csv …` flags the spawn-braked default, MaxForce vs curve, empty compressor, zero cut-out pressure, missing `CabControlName` targets and zero `MaxPermissibleSpeed` (Part 7).